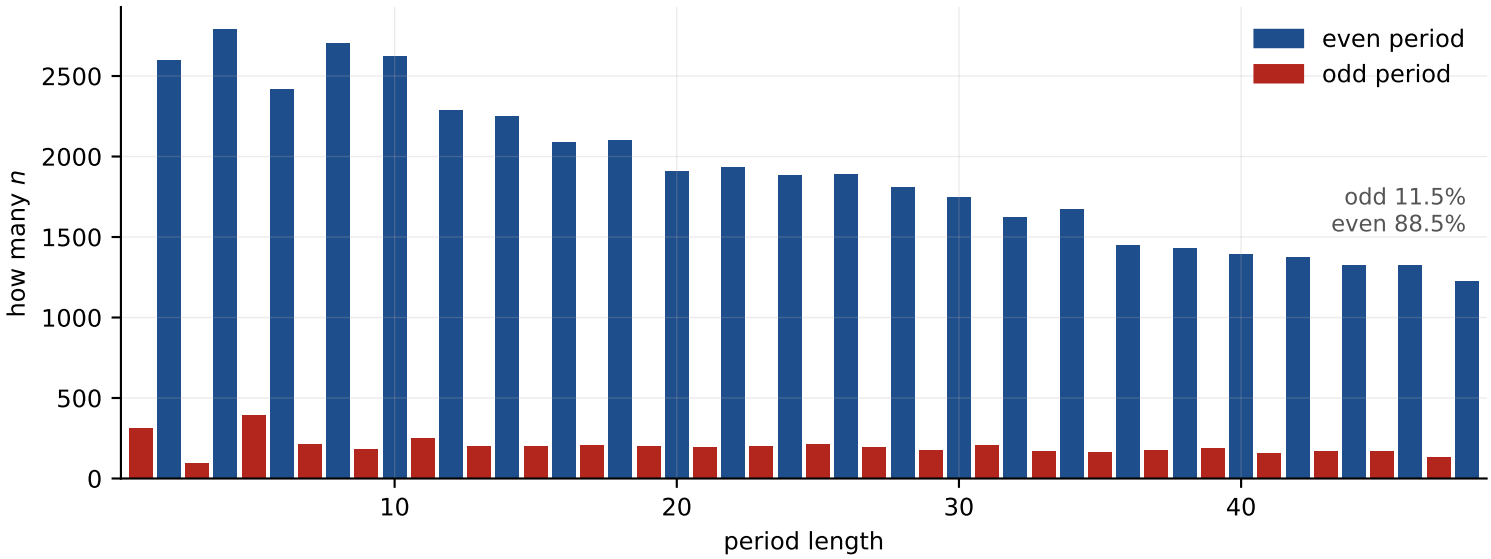
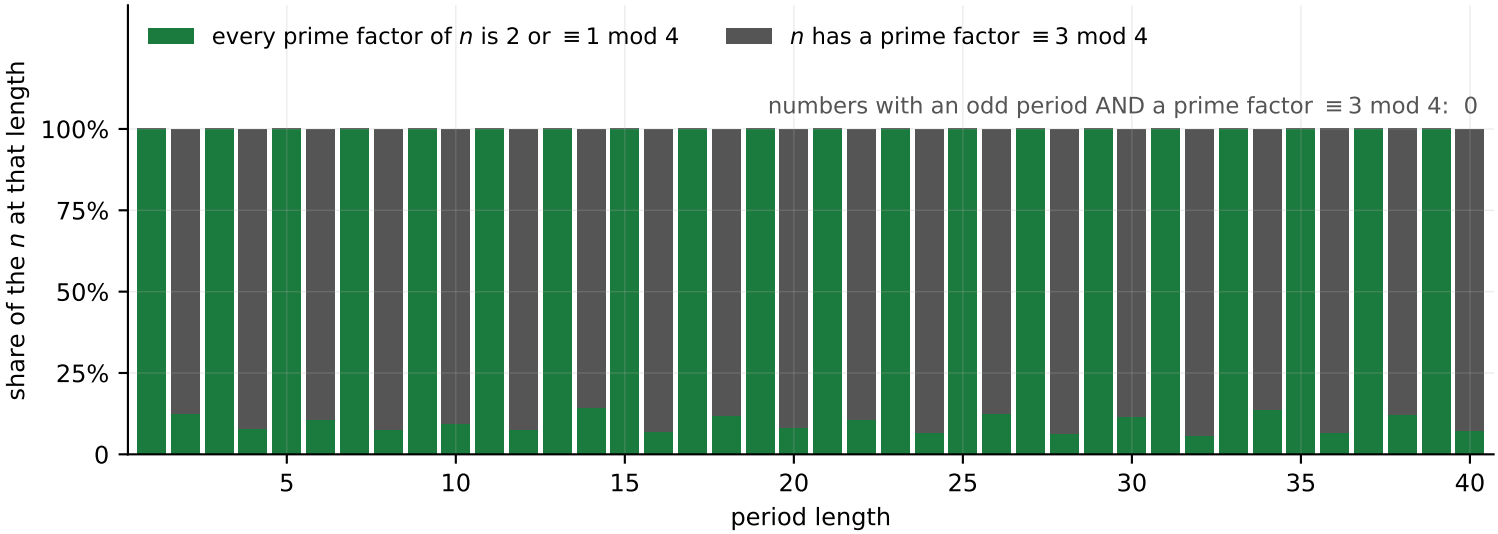
Period of the continued fraction of $\sqrt{n}$

every non-square n from 2 to 99999 (99,683 values)

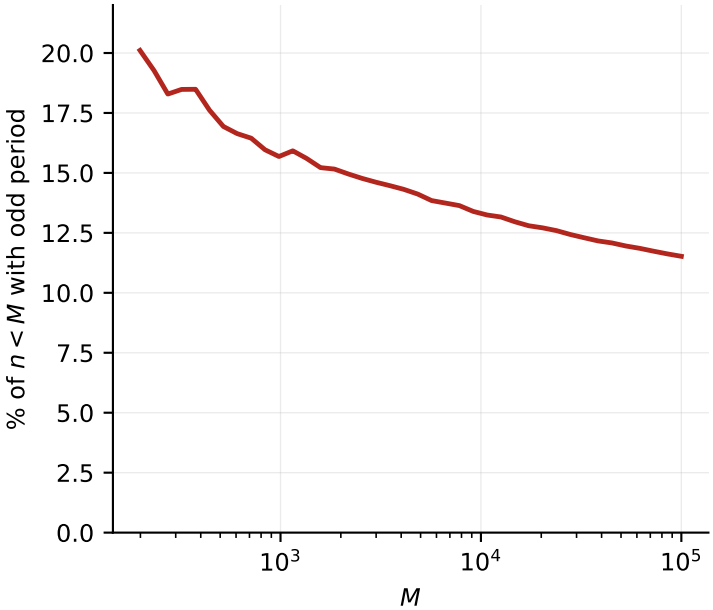
1. The comb: even periods are common, odd periods are rare



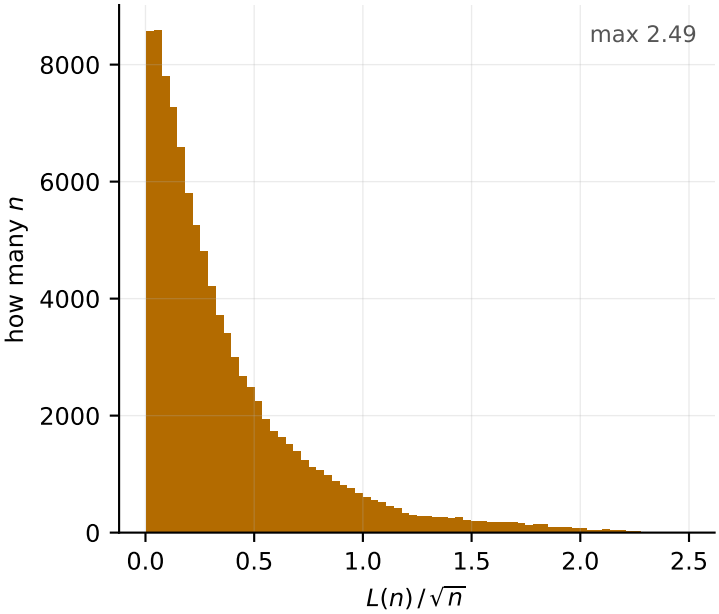
2. Why: every odd-period column is entirely green



3. The odd column keeps thinning

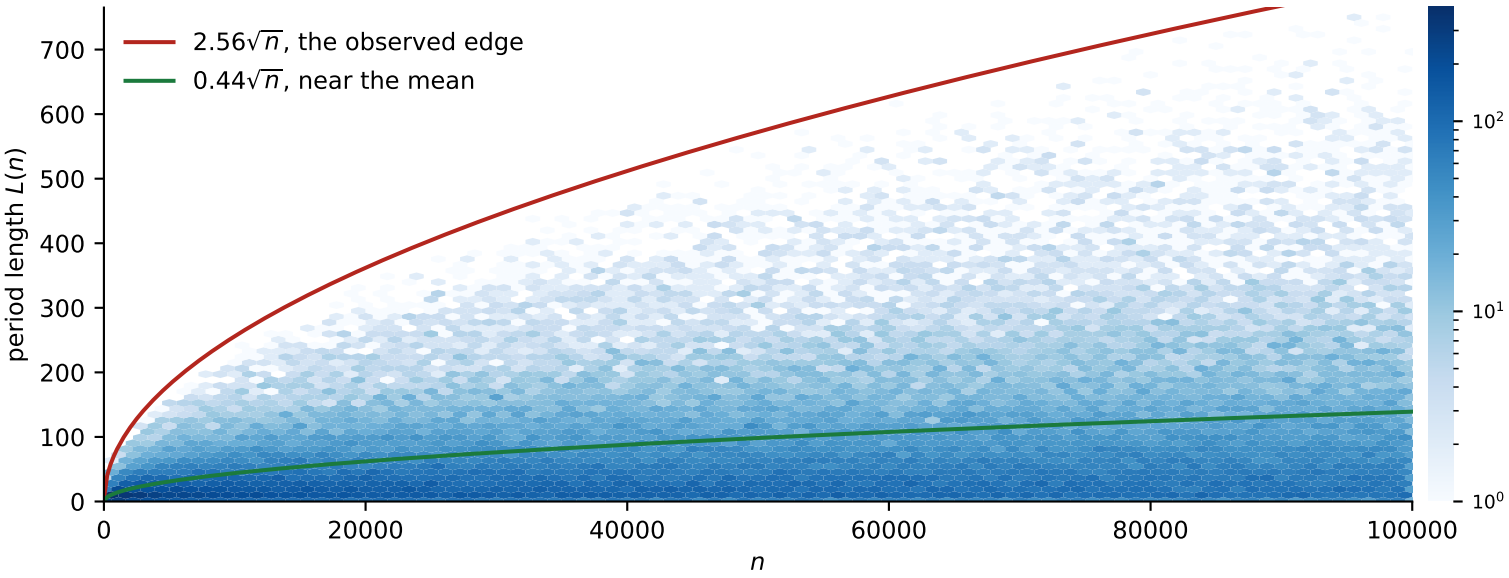


4. Divided by $\sqrt{n}$, the comb vanishes

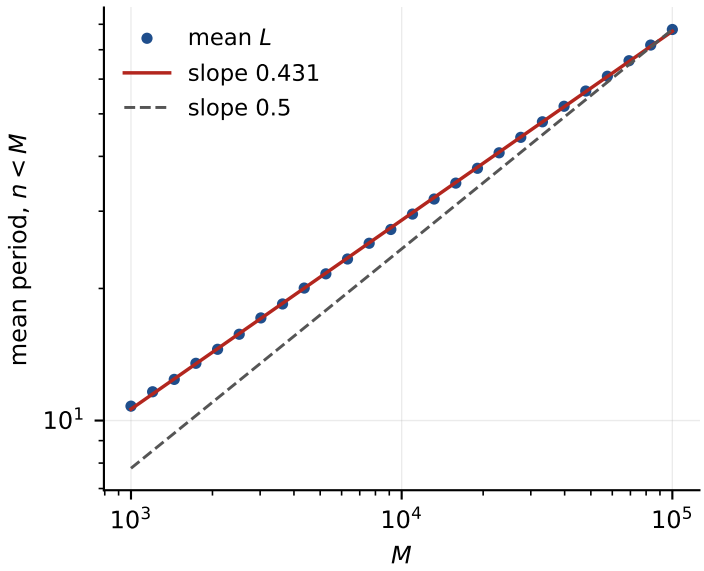


Growth, spread, and the edge

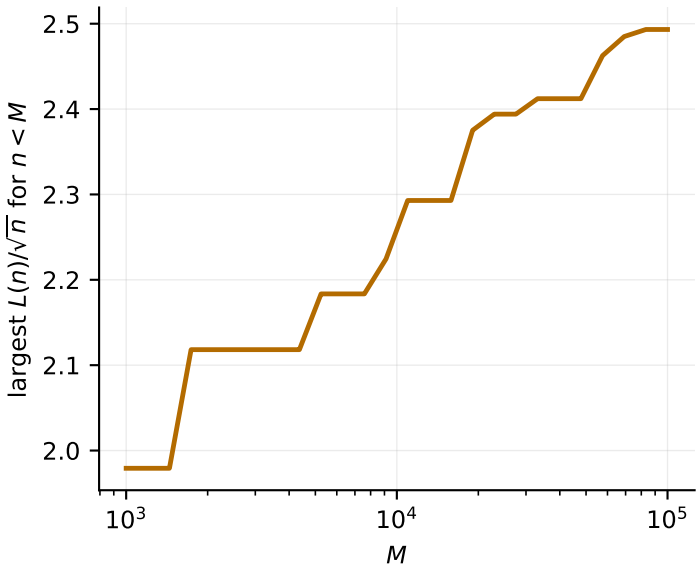
5. Every period, plotted against n (shading is how many, on a log scale)



6. The mean grows slower than $\sqrt{M}$



7. The right edge is not hard, it drifts



8. The full distribution, all period lengths

